2A 2

(b) $\Delta m = 226.0254 - (222.0176 + 4.0026) = 0.0052 \text{ u}$

1M

Energy released = $(0.0052)(931) = 4.84 \text{ (MeV)}$

1A 2

(c) Number of radium atoms in the source

$$N = N_A \left(\frac{1}{226} \right) \times (5 \times 10^{-6}) = (6.02 \times 10^{23}) \frac{1}{226} \times (5 \times 10^{-6}) = 1.33 \times 10^{16}$$

1A

$$\text{Activity } A = \frac{\ln 2}{t_{1/2}} \cdot N$$

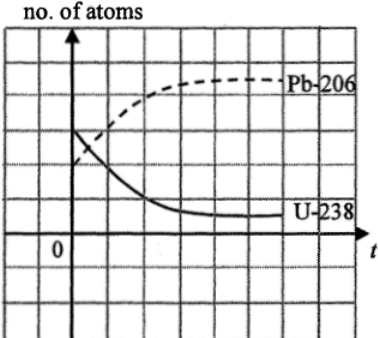
1M

$$= \frac{\ln 2}{1600 \times 365 \times 24 \times 3600} \cdot 1.33 \times 10^{16}$$

$$= 1.83 \times 10^5 \text{ (disintegrations per second, Bq)}$$

1A 3

10. (a) Mass deficit $= (2.014102 + 3.016049) \text{ u} - (4.002602 + 1.008665) \text{ u}$ $= 0.018884 \text{ u}$ Energy released $= 0.018884 \times 931 \text{ MeV}$ $= 17.58 \text{ (MeV)}$ Or Energy released $= 0.018884 \times 1.661 \times 10^{-27} \times c^2$ $= 2.823 \times 10^{-12} \text{ J or } 17.64 \text{ MeV}$	1M 1A 1A	
	2	
(b) (i) To overcome the (electrostatic) repulsion between the two (positive) nuclei and becomes electrical potential energy (of the two nuclei).	1A 1A 2	
(ii) High temperature enables them to have sufficient kinetic energy (to overcome electrical repulsion between the nuclei).	1A 1	
(iii) Kinetic energy becomes electrical potential energy $E_p = 2 \times \frac{1}{2} m (c_{\text{rms}})^2 = 2 \times \frac{3RT}{2N_A}$ $0.4 \text{ MeV} = 2 \times \left(\frac{3 \times 8.31 \times T}{2 \times 6.02 \times 10^{23}} \right)$ $T = 1.545 \times 10^9 \text{ K i.e. order of magnitude } 10^9 \text{ (K)}$	1M 1A	Accept without the factor '×2' 0.4 MeV = $6.4 \times 10^{-14} \text{ J}$
Alternatively: $E_p = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{10^{-15}} = 2 \times \frac{3RT}{2N_A}$ $T = 5.56 \times 10^9 \text{ K i.e. order of magnitude } 10^9 \text{ (K)}$	1M 1A	
	2	

9. (a) (4) $n_{\alpha} = 238 - 206 \Rightarrow n_{\alpha} = 8$ (2) $n_{\alpha} + (-1)n_{\beta} = 92 - 82 \Rightarrow n_{\beta} = 6$	1A 1A	
(b) (i) $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$ $\frac{3}{5} = \left(\frac{1}{2}\right)^{t/4.5 \times 10^9 \text{ yr}}$ Or $N = N_0 e^{-\lambda t}$ and $\lambda = \frac{\ln 2}{T_{1/2}}$ $\therefore t = 3.316 \times 10^9 \text{ years} \approx 3.3 \times 10^9 \text{ years}$	2 1M 1M 1A	
(ii) Answer in (i) is an underestimate. The original number of U-238 atoms should be greater. $\therefore$ the ratio $\frac{\text{present number of U-238 atoms}}{\text{original number of U-238 atoms}} = \frac{N_t}{N_0}$ is in fact smaller (than $\frac{3}{5}$), thus longer time should have been elapsed.	2 1A 1A	Accept 'more U decayed'.
(iii) 	2A 2	

10. (a)	${}_{84}^{210}\text{Po} \rightarrow {}_{82}^{206}\text{Pb} + {}_2^4\text{He}$	2A	
		2	
		1A	
		1A	
		2	
		1A	
		1	
		1M	
		1A	
		1M	
		1A	
		2	
(b)	The α -particles ionize the air particles. The ions neutralize the charges on the dust/photo or film surface.		
(c)	This is because α -particles have a range of only a few centimeters in air.		
(d)	Activity after 1 year = $\left(\frac{1}{2}\right)^{\frac{365}{138}}$ = 0.160 unit		
Alternative method:			
$A = A_0 e^{-\frac{\ln 2}{t_{1/2}} t}$			
$= 1 \times e^{-\frac{\ln 2}{138}(365)}$			
$= 0.160 \text{ unit}$			

Solution		Marks	Remarks
10. (a)	(i) $x = 3$	1A	
		1	
	(ii) More neutrons are produced in each fission for triggering further fissions, i.e. $x > 1$.	1A	
		1	
	(b) (i) $m = m_0 e^{-kt}$ $k = \frac{\ln 2}{t_{1/2}} (= 9.846 \times 10^{-10} \text{ yr}^{-1})$ $0.06 = m_0 e^{-\ln 2 \times \left[\frac{2 \times 10^9}{7.04 \times 10^8} \right]}$ $m_0 = 0.429882832 \text{ (kg)} \approx 0.430 \text{ (kg)}$	1M 1A	
		2	
	(ii) $\frac{0.430}{13.556 + 0.430} = 0.03073691 \approx 3.1 \% > 3\%$ Thus natural nuclear fission was possible.	1M/1A	
		1	
	(c) Underground water might run dry. <u>OR</u> Energy released by fission dries up the underground water.	1A 1A	
	Therefore, fission might stop without slow neutrons.	1A	
		2	

OR $0.06 = m_0 \left(\frac{1}{2}\right)^{\frac{2 \times 10^9}{7.04 \times 10^8}}$

10. (a) proton / ${}^1_1\text{H}$ / p / hydrogen nucleus	1A	Accept: proton / ${}^1_1\text{H}$ / p / 1_1p / H / hydrogen nucleus / hydrogen / hydrogen atom NOT accept: H_2
	1	
(b) Change in mass $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015)$ $= 0.0012 \text{ u}$ $\text{Energy} = 0.0012 \times 931$ $= 1.1172 \text{ (MeV)} \approx 1.12 \text{ (MeV)}$ (Or $= 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13} \text{ J} \approx 1.79 \times 10^{-19} \text{ MJ})$	1M	1M for Δm
	1A	Accept 1.10 to 1.12 (MeV) Answer should be in MeV, not J. MeV could be omitted $[\text{Energy} = 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13} \text{ J}$ $\approx 1.79 \times 10^{-19} \text{ MJ}$ $\approx 1.12 \text{ (MeV)}]$ Accept $(1.79 \text{ to } 1.80) \times 10^{-19} \text{ MJ}$
	2	
(c) By the conservation of momentum, as before the reaction occurs the α particle has momentum, the total momentum of the products (= momentum of the α particle) must also be non-zero, i.e. the total KE of the products must greater than 0, thus the α particle should have a larger KE.	1A	1 st 1A should mention the product(s) has/have momentum
	1A	2 nd 1A KE/speed of the products must greater than zero Accept: <u>kinetic</u> energy shared by other particles (NOT accept :energy shared.) NOT accept: against repulsion of nucleus / inelastic collisions / KE loss / consider binding energy [The kinetic energy of the α particle = energy calculated in (b) + total KE of the products]
	2	

10. (a)	Decrease in mass in the reaction (for two ${}^2_1\text{H}$) $\Delta m = (2 \times 2.014102) - (3.016029 + 1.008665)$ $= 4.028204 - 4.024694 = 0.003510 \text{ u}$ Maximum energy released by 1 mole of hydrogen nuclides $= \left(\frac{6.02 \times 10^{23}}{6420 \times 2} \right) \times (0.003510 \times 931 \text{ MeV})$ $= 1.532104 \times 10^{20} \text{ (MeV)} \approx 1.53 \times 10^{20} \text{ (MeV)}$	1M 1M 1A
(b)	${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_1\text{H} + {}^1_1\text{H}$ (or ${}^1_1\text{p}$)	1A 1
(c)	Any TWO: - Unlike fission, the fuel (hydrogen) for fusion is abundant and widely available. - Unlike fission, the products of fusion reactions are not radioactive. - Much larger amount of energy is produced by fusion (for equal mass of fuel).	2A 2

9. (a) (i)	alpha particle / α	1A	
		1	
(ii)	6 α -decays, 4 β -decays $4 n_{\alpha} = 232 - 208$ $2 n_{\alpha} + (-1) n_{\beta} = 90 - 82$	1A	
		1A	
	$n_{\alpha} = 6$ $n_{\beta} = 4$	2	
		1M	
(b)	$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} = \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$ $\text{Portion} = 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$ $= 4.9159 \times 10^{-10} \approx 4.9 \times 10^{-10} \text{ (2 sig. fig.)}$	1A	
		1A	
		2	
		1A	
(c) (i)	Aluminium body blocks alpha and/or beta particles emitted from thoriated glass. <u>Or</u> Short range of α particles	1A	
		1A	
		1	
		1A	
(ii)	Thorium-232 has an extremely long half-life. <u>Or</u> As part (b) suggests, only an extremely small fraction (4.9×10^{-10}) of thorium decaying in 10 years.	1A	
		1A	
		1	
		1A	

$$N = N_0 e^{-\lambda t} \quad \text{and} \quad \lambda = \frac{\ln 2}{t_{1/2}}$$

$$\lambda = \frac{\ln 2}{t_{1/2}} = 4.9159 \times 10^{-11} \text{ yr}^{-1}$$

$$1 - \frac{N}{N_0} = 1 - e^{-\frac{\ln 2}{t_{1/2}} t}$$

$$= 1 - e^{-\frac{\ln 2}{1.41 \times 10^{10}} (10)}$$

$$\approx 4.9 \times 10^{-10}$$